

PREDEAL, Romania

WMO: 153020

Lat: 45.5064N

Lon: 25.5836E

Elev: 3580

StdP: 12.89

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.4	8.3	0.5	6.4	7.3	4.4	7.9	11.1	18.7	28.9	16.5	24.6	1.9	320	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	77.8	62.6	75.0	61.4	72.3	60.3	64.8	73.8	63.3	71.5	62.0	69.6	3.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.0	95.1	67.2	60.5	90.1	65.9	59.1	85.6	64.9	31.9	74.1	30.6	71.4	29.6	69.8	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.6	11.6	9.4	-2.1	82.4	3.8	3.1	-4.8	84.6	-7.0	86.4	-9.2	88.1	-11.9	90.4		
			-2.5	68.4	3.8	1.4	-5.3	69.4	-7.5	70.3	-9.7	71.1	-12.5	72.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.1	25.5	28.9	32.6	42.0	50.6	57.7	59.9	60.0	52.2	43.9	35.9	27.4
	DBStd	14.45	8.82	8.71	7.94	8.20	6.00	5.49	4.90	5.15	7.58	7.70	8.68	8.34
	HDD50	3631	760	590	540	259	65	7	1	1	67	217	424	700
	HDD65	8009	1225	1010	1004	691	445	224	167	168	385	653	872	1165
	CDD50	1126	0	0	0	18	84	240	308	311	134	29	2	0
	CDD65	30	0	0	0	0	0	6	9	13	2	0	0	0
	CDH74	352	0	0	0	1	4	62	108	146	30	1	0	0
	CDH80	31	0	0	0	0	0	6	6	16	3	0	0	0
	WSAvg	3.2	3.1	3.4	3.6	3.5	3.3	3.3	3.2	2.8	3.2	3.1	3.2	3.1
	PrecAvg	37.4	1.9	1.9	1.8	2.9	4.6	5.6	5.4	3.9	2.8	2.2	2.2	2.2
Precipitation	PrecMax	47.7	4.5	4.1	5.4	4.9	9.2	10.7	12.0	6.9	7.4	7.4	4.9	6.8
	PrecMin	24.3	0.4	0.1	0.1	0.6	1.2	1.5	1.5	0.7	0.2	0.4	0.7	0.1
	PrecStd	6.1	1.1	0.9	1.3	1.1	2.0	2.3	2.5	1.8	2.0	1.7	1.0	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.0	55.2	59.7	68.3	73.6	79.4	80.4	82.2	77.5	68.8	59.9	49.9
		MCWB	38.9	44.4	45.6	51.8	55.7	64.0	64.0	65.1	61.9	54.2	49.0	42.7
	2%	DB	43.4	48.8	53.7	65.0	69.6	75.2	77.1	77.8	72.6	65.0	55.8	43.9
		MCWB	38.4	40.6	43.2	50.9	55.3	61.6	63.0	62.4	58.6	53.1	47.8	39.7
	5%	DB	39.6	44.8	49.0	60.1	66.5	72.2	74.2	74.6	68.8	60.8	52.2	40.9
		MCWB	35.7	38.2	40.5	48.5	54.4	60.4	61.9	61.1	57.5	51.5	45.6	37.2
	10%	DB	36.0	41.0	44.6	55.4	63.0	69.4	71.0	71.3	65.2	55.9	48.4	37.5
		MCWB	33.8	36.2	38.2	46.4	53.0	59.5	61.1	60.6	56.0	49.4	44.0	35.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.3	45.0	47.2	53.9	59.6	66.4	66.6	66.9	62.9	57.6	50.6	43.3
		MCDB	45.8	53.6	57.9	64.9	67.3	76.1	75.8	77.7	75.0	63.3	56.8	46.1
	2%	WB	39.1	41.3	43.6	51.6	57.3	63.6	64.6	64.8	60.2	54.6	48.6	40.6
		MCDB	42.9	47.3	52.2	62.7	65.9	71.3	72.9	74.5	69.1	62.1	54.6	44.6
	5%	WB	36.3	39.1	41.4	49.6	55.6	61.9	63.2	63.0	58.4	52.3	46.7	37.5
		MCDB	38.8	43.4	48.0	58.5	64.3	69.4	70.9	71.4	66.4	59.3	51.9	40.3
	10%	WB	34.0	36.9	38.9	47.4	53.9	60.5	61.9	61.6	56.8	50.3	44.4	35.5
		MCDB	35.8	40.6	43.6	54.0	61.1	67.8	69.0	69.2	63.8	55.5	47.9	37.3
Mean Daily Temperature Range	5% DB	MDBR	12.5	14.1	14.2	16.9	17.8	17.8	17.0	18.2	16.6	15.7	12.9	11.2
		MCDBR	17.1	19.8	22.6	25.5	24.5	22.9	22.9	23.6	22.9	23.4	19.5	15.1
		MCWBR	11.4	12.4	13.0	13.0	12.2	11.2	10.6	10.7	10.8	12.9	11.4	10.3
	5% WB	MCDBR	15.7	18.2	20.5	23.1	21.6	20.8	20.2	21.1	20.4	21.9	17.2	13.6
		MCWBR	11.1	12.0	12.7	12.3	11.6	11.1	10.5	10.9	10.3	12.7	10.5	9.9
Clear-Sky Solar Irradiance	taub		0.277	0.303	0.341	0.393	0.377	0.378	0.391	0.403	0.364	0.331	0.308	0.280
	taud		2.368	2.307	2.296	2.216	2.320	2.339	2.304	2.268	2.359	2.448	2.465	2.415
	Ebn at Noon		265	277	281	274	283	282	276	268	270	265	252	252
	Edh at Noon		23	30	36	42	39	39	40	40	33	26	22	20
All-Sky Solar Radiation	RadAvg		441	701	1040	1357	1656	1789	1820	1648	1209	835	487	361
	RadStd		45	88	109	150	162	162	157	138	117	106	49	43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)	+1.03	N/A	N/A	+1.19	N/A	N/A	-218	-299	+126	+41

Nomenclature: See separate page